

## Solving analytical queries on Redshift Cluster

Here, you have to write the query used for solving the question and the screenshots of the table which is outputted after the query is run on the AWS Redshift Query editor UI.

### 1. Top 10 ATMs where most transactions are in the 'inactive' state

```
select a.atm_number, a.atm_manufacturer, l.location, count(trans_id) as
total_transaction_count, sum(case when f.atm_status = 'Inactive'then 1 else 0 end) as
inactive_count, (inactive_count/total_transaction_count)*100 as count_percent from
atm_info.FACT_ATM_TRANS f join atm_info.DIM_ATM a on a.atm_id = f.atm_id join
atm_info.DIM_LOCATION l on a.atm_location_id = l.location_id group by a.atm_number,
a.atm_manufacturer, l.location having count_percent > 50 order by inactive_count desc limit 10;
```

Completed, started on March 08, 2025 at 23:17:35  
ELAPSED TIME: 00 m 14 s

Rows returned (10) Export ▼

Search rows

| atm_number | atm_manufacturer | location                | total_transaction_count | inactive_count | count_percent |
|------------|------------------|-------------------------|-------------------------|----------------|---------------|
| 16         | NCR              | Skive                   | 44043                   | 44043          | 100           |
| 30         | NCR              | Nykøbing Mors           | 33982                   | 33982          | 100           |
| 4          | NCR              | Svovgerslev             | 33725                   | 33725          | 100           |
| 88         | NCR              | Storcenter indg. A      | 32183                   | 32183          | 100           |
| 12         | NCR              | Århus                   | 30883                   | 30883          | 100           |
| 72         | NCR              | Daglig Brugsen<br>Århus | 27361                   | 27361          | 100           |
| 50         | NCR              | Aarhus                  | 23416                   | 23416          | 100           |
| 29         | NCR              | Skelagervej 15          | 20773                   | 20773          | 100           |
| 107        | Diebold Nixdorf  | Kolding                 | 18297                   | 18297          | 100           |
| 102        | NCR              | Aalborg Storcenter Afd  | 17164                   | 17164          | 100           |

## 2. Number of ATM failures corresponding to the different weather conditions recorded at the time of the transactions

```
select weather_main, count(trans_id) as total_transaction_count, sum(case when atm_status = 'Inactive' then 1 else 0 end) as inactive_count, case when coalesce(inactive_count, 0) != 0 then trunc((cast(inactive_count as numeric(10,4))/total_transaction_count)*100, 2) else 0.0000 end as inactive_count_percent from atm_info.FACT_ATM_TRANS where weather_main != " group by weather_main order by inactive_count_percent desc limit 10;
```

| Query Execution                                                             |                         |                |                        | Execution | Data | Visualize |
|-----------------------------------------------------------------------------|-------------------------|----------------|------------------------|-----------|------|-----------|
| Completed, started on March 08, 2025 at 23:20:16<br>ELAPSED TIME: 00 m 13 s |                         |                |                        |           |      |           |
| Rows returned (10)                                                          |                         |                |                        | Export ▼  |      |           |
| Q Search rows                                                               |                         |                |                        | < 1 > ⚙   |      |           |
| weather_main                                                                | total_transaction_count | inactive_count | inactive_count_percent |           |      |           |
| Snow                                                                        | 23405                   | 4813           | 20.5600                |           |      |           |
| Fog                                                                         | 18174                   | 3729           | 20.5100                |           |      |           |
| Clouds                                                                      | 1181901                 | 194027         | 16.4100                |           |      |           |
| Rain                                                                        | 545135                  | 86017          | 15.7700                |           |      |           |
| Clear                                                                       | 543949                  | 85531          | 15.7200                |           |      |           |
| Mist                                                                        | 82801                   | 12864          | 15.5300                |           |      |           |
| Thunderstorm                                                                | 2549                    | 361            | 14.1600                |           |      |           |
| Drizzle                                                                     | 62530                   | 8670           | 13.8600                |           |      |           |
| TORNADO                                                                     | 38                      | 1              | 2.6300                 |           |      |           |
| Haze                                                                        | 3                       | 0              | 0.0000                 |           |      |           |

### 3. Top 10 ATMs with the most number of transactions throughout the year

```
select a.atm_number, a.atm_manufacturer, l.location, count(trans_id) as total_transaction_count
from atm_info.FACT_ATM_TRANS f join atm_info.DIM_ATM a on a.atm_id = f.atm_id join
atm_info.DIM_LOCATION l on a.atm_location_id = l.location_id group by a.atm_number,
a.atm_manufacturer, l.location order by total_transaction_count desc limit 10;
```

Query [607279](#)

Execution

Data

Visualize

Completed, started on March 08, 2025 at 23:22:02

ELAPSED TIME: 00 m 15 s

Rows returned (10)

Export

Search rows

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| atm_number | atm_manufacturer | location            | total_transaction_count |
|------------|------------------|---------------------|-------------------------|
| 39         | NCR              | Svenstrup           | 55380                   |
| 95         | NCR              | Intern K&A, benhavn | 54211                   |
| 110        | Diebold Nixdorf  | Holb&f&k            | 53794                   |
| 24         | NCR              | Hobro               | 53378                   |
| 45         | NCR              | Abildgaard          | 53198                   |
| 16         | NCR              | Skive               | 44043                   |
| 40         | Diebold Nixdorf  | Frederikshavn       | 43767                   |
| 1          | NCR              | N&f&stved           | 42787                   |
| 41         | Diebold Nixdorf  | Skagen              | 42732                   |
| 48         | Diebold Nixdorf  | Br&f&, nderslev     | 42493                   |

#### 4. Number of overall ATM transactions going inactive per month for each month

```
select year, month, count(trans_id) as total_transaction_count, sum(case when atm_status = 'Inactive' then 1 else 0 end) as inactive_count, case when coalesce(inactive_count, 0) != 0 then trunc((cast(inactive_count as numeric(10,4))/total_transaction_count)*100, 2) else 0.0000 end as inactive_count_percent from atm_info.DIM_DATE d join atm_info.FACT_ATM_TRANS f on f.date_id = d.date_id group by year, month order by year, month;
```

Query [607309](#)

Completed, started on March 08, 2025 at 23:24:40  
ELAPSED TIME: 00 m 20 s

Execution

Data

Visualize

Rows returned (12)

Export

Search rows

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| year | month    | total_transaction_count | inactive_count | inactive_count_percent |
|------|----------|-------------------------|----------------|------------------------|
| 2017 | April    | 208208                  | 34644          | 16.6300                |
| 2017 | August   | 213300                  | 38101          | 17.8600                |
| 2017 | December | 218345                  | 29349          | 13.4400                |
| 2017 | February | 184147                  | 33391          | 18.1300                |
| 2017 | January  | 185816                  | 33848          | 18.2100                |
| 2017 | July     | 222337                  | 38945          | 17.5100                |
| 2017 | June     | 195055                  | 28657          | 14.6900                |
| 2017 | March    | 213263                  | 37819          | 17.7300                |
| 2017 | May      | 213389                  | 31730          | 14.8600                |
| 2017 | November | 201264                  | 27864          | 13.8400                |

## 5. Top 10 ATMs with the highest total withdrawn amount throughout the year

```
select a.atm_number, a.atm_manufacturer, l.location, sum(transaction_amount) as
total_transaction_amount from atm_info.FACT_ATM_TRANS f join atm_info.DIM_ATM a on
a.atm_id = f.atm_id join atm_info.DIM_LOCATION l on a.atm_location_id = l.location_id group
by a.atm_number, a.atm_manufacturer, l.location order by total_transaction_amount desc limit
10;
```

Query [607336](#)

Execution

Data

Visualize

Completed, started on March 08, 2025 at 23:26:50

ELAPSED TIME: 00 m 14 s

Rows returned (10)

Export

Search rows

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| atm_number | atm_manufacturer | location             | total_transaction_amount |
|------------|------------------|----------------------|--------------------------|
| 39         | NCR              | Svenstrup            | 277097637                |
| 95         | NCR              | Intern K&f&, benhavn | 271008803                |
| 24         | NCR              | Hobro                | 268289882                |
| 110        | Diebold Nixdorf  | Holb&f&lk            | 267379103                |
| 45         | NCR              | Abildgaard           | 265639616                |
| 16         | NCR              | Skive                | 220677013                |
| 40         | Diebold Nixdorf  | Frederikshavn        | 219812287                |
| 41         | Diebold Nixdorf  | Skagen               | 214127315                |
| 1          | NCR              | N&f&stved            | 213721117                |
| 48         | Diebold Nixdorf  | Br&f&, nderslev      | 212883099                |

## 6. Number of failed ATM transactions across various card types

```
select card_type, count(trans_id) as total_transaction_count, sum(case when atm_status =
'Inactive' then 1 else 0 end) as inactive_count, case when coalesce(inactive_count, 0) != 0 then
trunc((cast(inactive_count as numeric(10,4))/total_transaction_count)*100, 2) else 0.0000 end
as inactive_count_percent from atm_info.DIM_CARD_TYPE c join
atm_info.FACT_ATM_TRANS f on f.card_type_id = c.card_type_id group by card_type order by
inactive_count_percent desc limit 10;
```

Query [607365](#)

Execution
Data
Visualize

Completed, started on March 08, 2025 at 23:28:26  
ELAPSED TIME: 00 m 13 s

Rows returned (10)

Export

Search rows

| card_type            | total_transaction_count | inactive_count | inactive_count_percent |
|----------------------|-------------------------|----------------|------------------------|
| Mastercard - on-us   | 458226                  | 86000          | 18.7600                |
| VISA                 | 170828                  | 30713          | 17.9700                |
| Dankort - on-us      | 143813                  | 24680          | 17.1600                |
| CIRRUS               | 17362                   | 2953           | 17.0000                |
| HÃ¶fÃ¶vekort - on-us | 62487                   | 10331          | 16.5300                |
| Dankort              | 28581                   | 4557           | 15.9400                |
| MasterCard           | 400507                  | 63482          | 15.8500                |
| Visa Dankort - on-us | 748805                  | 112972         | 15.0800                |
| HÃ¶fÃ¶vekort         | 8459                    | 1208           | 14.2800                |
| Visa Dankort         | 427840                  | 60547          | 14.1500                |

**7. Number of transactions happening on an ATM on weekdays and on weekends throughout the year. Order this by the ATM\_number, ATM\_manufacturer, location, weekend\_flag and then total\_transaction\_count**

```
select atm_number, atm_manufacturer, location, case when weekday in ('Sunday','Saturday')
then 1 else 0 end as weekend_flag, count(trans_id) as total_transaction_count from
atm_info.FACT_ATM_TRANS f join atm_info.DIM_ATM a on a.atm_id = f.atm_id join
atm_info.DIM_LOCATION l on a.atm_location_id = l.location_id join atm_info.DIM_DATE d on
f.date_id = d.date_id group by atm_number, atm_manufacturer, location,weekend_flag order by
atm_number, atm_manufacturer, location,weekend_flag, total_transaction_count limit 10;
```

Completed, started on March 08, 2025 at 23:30:08  
ELAPSED TIME: 00 m 14 s

| Rows returned (10)                       |                  |                        |              |                         |  | Export ▼ |
|------------------------------------------|------------------|------------------------|--------------|-------------------------|--|----------|
| <input type="text" value="Search rows"/> |                  |                        |              |                         |  | < 1 > ⚙️ |
| atm_number                               | atm_manufacturer | location               | weekend_flag | total_transaction_count |  |          |
| 1                                        | NCR              | NÅfÅ\stved             | 0            | 31568                   |  |          |
| 1                                        | NCR              | NÅfÅ\stved             | 1            | 11219                   |  |          |
| 10                                       | NCR              | NÅfÅ ,rresundby        | 0            | 3483                    |  |          |
| 10                                       | NCR              | NÅfÅ ,rresundby        | 1            | 847                     |  |          |
| 100                                      | NCR              | Intern Skive           | 0            | 15981                   |  |          |
| 100                                      | NCR              | Intern Skive           | 1            | 1832                    |  |          |
| 100                                      | NCR              | Skive                  | 0            | 14006                   |  |          |
| 100                                      | NCR              | Skive                  | 1            | 4735                    |  |          |
| 102                                      | NCR              | Aalborg Storcenter Afd | 0            | 12392                   |  |          |
| 102                                      | NCR              | Aalborg Storcenter Afd | 1            | 4772                    |  |          |

## 8. Most active day in each ATMs from location "Vejgaard"

with cte as ( select atm\_number, atm\_manufacturer, location, weekday, count(trans\_id) as total\_transaction\_count, row\_number() over (partition by atm\_number, atm\_manufacturer, location order by count(trans\_id) desc) as rank from atm\_info.FACT\_ATM\_TRANS f join atm\_info.DIM\_ATM a on f.atm\_id = a.atm\_id join atm\_info.DIM\_DATE d on f.date\_id = d.date\_id join atm\_info.DIM\_LOCATION l on l.location\_id = f.weather\_loc\_id where location = 'Vejgaard' group by atm\_number, atm\_manufacturer, location, weekday ) select atm\_number, atm\_manufacturer, location, weekday, total\_transaction\_count from cte where rank = 1;

 Completed, started on March 09, 2025 at 14:19:33  
ELAPSED TIME: 00 m 02 s

Rows returned (2)

Export

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| atm_number | atm_manufacturer | location | weekday | total_transaction_count |
|------------|------------------|----------|---------|-------------------------|
| 2          | NCR              | Vejgaard | Friday  | 5760                    |
| 103        | Diebold Nixdorf  | Vejgaard | Friday  | 4127                    |